

Chapter 2: The first law of thermodynamics

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Review

- ✓ Consider a body, initially in the **state A**, which performs work **w**, absorbs heat **q**, and, as a consequence, moves to the **state B**.
- ✓ The absorption of heat q *increases* the internal energy of the body by the amount q , and the performance of work w by the body *decreases* its internal energy by the amount w . Thus the total change in the internal energy of the body is:

$$\Delta U = U_B - U_A = q - w$$

$$dU = \delta q - \delta w$$

The first law of thermodynamics

Review

- ✓ The heat capacity, C , of a system is the ratio of the heat added to or withdrawn from the system to the resultant change in the temperature of the system.

$$C = \frac{q}{\Delta T}$$

$$C = \frac{\delta q}{\delta T}$$

$$C_V = \left(\frac{\delta q}{dT} \right)_V$$

The heat capacity at constant volume

$$C_P = \left(\frac{\delta q}{dT} \right)_P$$

The heat capacity at constant pressure

molar heat capacity

$$C_V = \left(\frac{\delta q}{dT} \right)_V = \left(\frac{dU}{dT} \right)_V$$

$$dU = C_V dT$$

$$C_V = n c_V$$

$$C_P = \left(\frac{\delta q}{dT} \right)_P = \left(\frac{dH}{dT} \right)_P$$

$$dH = C_P dT$$

$$C_P = n c_P$$

Review

✓It is to be expected that, for any substance, C_p will be of greater magnitude than C_v .

$$c_P = \left(\frac{\delta q}{dT} \right)_P = \left(\frac{dH}{dT} \right)_P = \left(\frac{\partial U}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P$$

$$c_V = \left(\frac{\delta q}{dT} \right)_V = \left(\frac{dU}{dT} \right)_V = \left(\frac{\partial U}{\partial T} \right)_V$$

$$c_P - c_V = \left(\frac{\partial U}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P - \left(\frac{\partial U}{\partial T} \right)_V$$

Review

$$U = U(V, T) \Rightarrow dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT$$

$$\left(\frac{\partial U}{\partial T} \right)_P = \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P + \left(\frac{\partial U}{\partial T} \right)_V$$

$$c_P - c_V = \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P + \left(\frac{\partial U}{\partial T} \right)_V + P \left(\frac{\partial V}{\partial T} \right)_P - \left(\frac{\partial U}{\partial T} \right)_V$$

$$= \left(\frac{\partial V}{\partial T} \right)_P \left[P + \left(\frac{\partial U}{\partial V} \right)_T \right]$$

The work done by the system per degree rise in temperature in expanding against the constant external pressure P acting on the system.

The work done per degree rise in temperature in expanding against the internal cohesive forces acting between the constituent particles of the substance.

Review

- ✓ An ideal gas is a gas consisting of non-interacting particles, and hence, the atoms or molecules of an ideal gas can be separated from one another without the expenditure of work. Thus, for an ideal gas.

$$\left(\frac{\partial V}{\partial T}\right)_P = \alpha V \neq 0 \quad \Rightarrow \quad \left(\frac{\partial V}{\partial T}\right)_P \left(\frac{\partial U}{\partial V}\right)_T = 0$$

$$\Rightarrow \quad \left(\frac{\partial U}{\partial V}\right)_T = 0$$

- In real gases the internal pressure contribution is very much smaller in magnitude than the external pressure contribution
- In liquids and solids, in which the interatomic forces are considerable, the work done in expanding the system against the external pressure is insignificant in comparison with the work done against the internal pressure. Thus for liquids and solids the term is very large.

Review

- ✓ ΔU in real gas is not only related to the temperature, but also is related to the volume (or pressure).

$$dU = \left(\frac{\partial U}{\partial T} \right)_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$

- ✓ Because of the interaction between molecules of real gas, the change in thermodynamic energy can be measured by the energy dissipated against the intermolecular attraction during isothermal expansion.

$\left(\frac{\partial U}{\partial V} \right)_T$ is called the internal pressure, that is: $P_{Inter} = \left(\frac{\partial U}{\partial V} \right)_T$

For isothermal condition, $dU = P_{Inter} dV$

2.6 Heat capacity

- ◆ Heat capacities can be used in conjunction with the First Law of Thermodynamics to calculate enthalpy changes of reactions as a function of temperature. Consider the solidification of 1 mole of a liquid to a solid at its equilibrium freezing temperature T_2 .

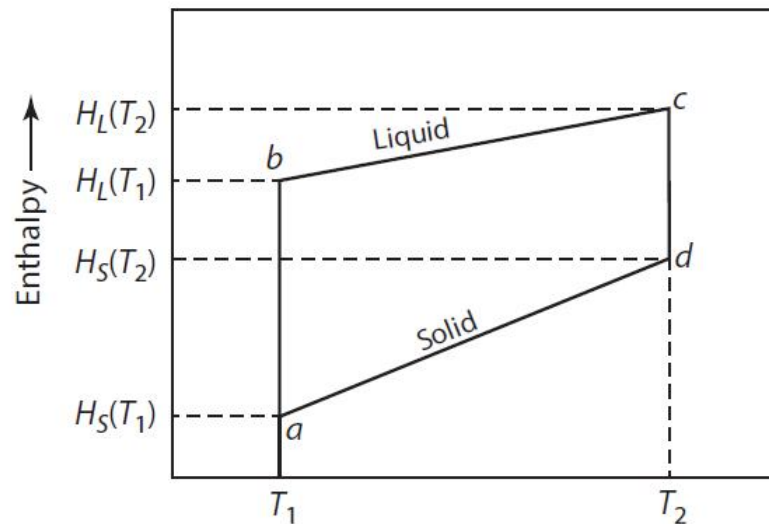


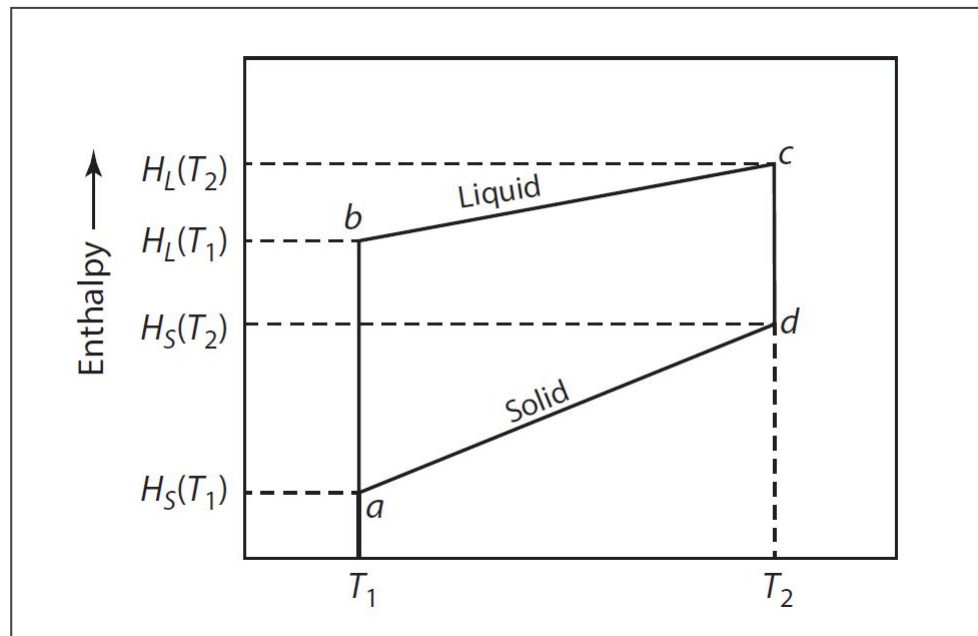
Figure 2.3 Enthalpy vs. temperature plots for 1 mole of a solid and liquid, showing that the enthalpy of transformation depends on temperature if the heat capacities differ. T_2 is the equilibrium solidification temperature.

2.6 Heat capacity

- The value of its enthalpy of freezing (or solidification) is seen to be.

$$H_S(T_2) - H_L(T_2) < 0$$

- Since thermal energy leaves the substance and enters the surroundings (exothermic). This value is the negative of the length of the cord $\overline{cd}$ in Figure 2.3. If the liquid is **supercooled** to a temperature of T_1 , the enthalpy change for freezing is seen now to be $H_S(T_1) - H_L(T_1)$.



2.6 Heat capacity

- The enthalpy of the liquid at T_1 .

$$H_L(T_1) = H_L(T_2) + \int_{T_2}^{T_1} c_P^L dT = H_L(T_2) + c_P^L (T_1 - T_2)$$

- The enthalpy of the solid at T_1 .

$$H_S(T_1) = H_S(T_2) + \int_{T_2}^{T_1} c_P^S dT = H_S(T_2) + c_P^S (T_1 - T_2)$$

Therefore

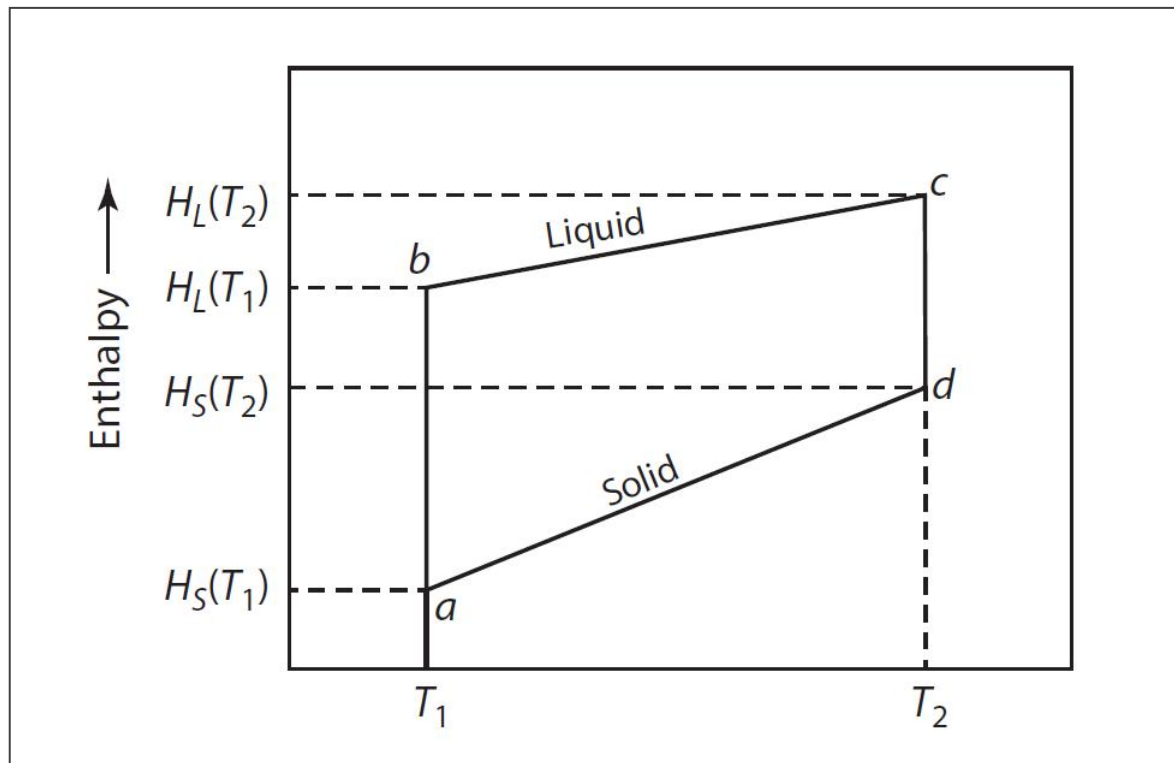
$$H_S(T_1) - H_L(T_1) = H_S(T_2) - H_L(T_2) + \int_{T_2}^{T_1} (c_P^S - c_P^L) dT$$

This value is more negative than that at T_2 because the temperature dependence of the enthalpy of the solid is greater than that of the liquid.

2.6 Heat capacity

The change in the enthalpy as a function of temperature can be written as

$$\begin{aligned} H_S(T_1) - H_L(T_1) = \\ H_S(T_2) - H_L(T_2) + \int_{T_2}^{T_1} (c_P^S - c_P^L) dT \end{aligned} \quad \Rightarrow \quad \left(\frac{\partial \Delta H}{\partial T} \right)_P = \Delta c_P$$



2.6 Heat capacity

Supercooling/Undercooling

- Supercooling, also known as undercooling, is the process of lowering the temperature of a liquid or a gas below its freezing point without it becoming a solid.
- A liquid crossing its standard freezing point will crystalize in the presence of a seed crystal or **nucleus** around which a crystal structure can form creating a solid. Lacking any such nuclei, the liquid phase can be maintained all the way down to the temperature at which crystal homogeneous **nucleation** occurs.
- Homogeneous nucleation can occur above the glass transition temperature, but if homogeneous nucleation has not occurred above that temperature, an **amorphous** (non-crystalline) solid will form.

2.7 Reversible adiabatic processes

- ◆ For a reversible process during which the state of the gas is changed, the state of the gas never leaves the equilibrium surface. Consequently, during a reversible process, the gas passes through a continuum of equilibrium states, and the work w is given by the integral $\int P dV$ only if the process is conducted reversibly. In an adiabatic process $q=0$, and thus, from the First Law, $dU=-\delta w$.
- Consider a system comprising one mole of an ideal gas.

$$dU = c_V dT$$

- For a reversible adiabatic process

$$\delta w = P dV$$

$$C_V dT = -P dV$$

2.7 Reversible adiabatic processes

$$C_V dT = -PdV$$

As the system is one mole of ideal gas, then $P=RT/V$ and hence

$$C_V dT = -PdV = -\frac{RT}{V} dV$$

$$C_V \ln\left(\frac{T_2}{T_1}\right) = R \ln\left(\frac{V_1}{V_2}\right) \quad \left(\frac{T_2}{T_1}\right)^{C_V} = \left(\frac{V_1}{V_2}\right)^R$$

$$\left(\frac{T_2}{T_1}\right) = \left(\frac{V_1}{V_2}\right)^{\frac{R}{C_V}} = \left(\frac{V_1}{V_2}\right)^{\frac{C_P - C_V}{C_V}} = \left(\frac{V_1}{V_2}\right)^{\gamma - 1}$$

2.7 Reversible adiabatic processes

$$\frac{T_2}{T_1} = \frac{P_2 V_2}{P_1 V_1} = \left(\frac{V_1}{V_2} \right)^{\gamma-1}$$

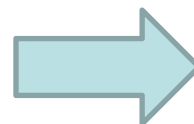
$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^{\gamma} \qquad P_2 V_2^{\gamma} = P_1 V_1^{\gamma} = \text{constant}$$

- This is the relationship between the pressure and the volume of an ideal gas under-going a reversible adiabatic process.

- For an ideal gas,

$$c_P = \frac{5}{2} R$$

$$c_V = \frac{3}{2} R$$



$$\gamma = \frac{5}{3}$$

2.8 Reversible isothermal pressure or volume changes of an ideal gas

$$dU = \delta q - \delta w$$

- As $dT=0$ (isothermal process), then $dU=0$. Therefore $\delta w=\delta q=PdV=RTdV/V$ per mole of gas. Integrating between the states 1 and 2 gives.

$$w = q = RT \ln \frac{V_2}{V_1} = RT \ln \frac{P_1}{P_2}$$

- Thus, for an ideal gas, an isothermal process is the one of constant internal energy during which the work done by the system equals the heat absorbed by the system.

2.8 Reversible isothermal pressure or volume changes of an ideal gas

- The work done by the reversible isothermal process (which is equal to the area under the curve) exceeds that done by the reversible adiabatic process.
- This difference is due to the fact that during the isothermal process heat is absorbed by the system in order to maintain the temperature constant, whereas during the adiabatic process no heat is admitted to the system.
- During the isothermal expansion, the internal energy of the gas remains constant, and during the adiabatic expansion the internal energy decreases by an amount equal to the work done by the gas.

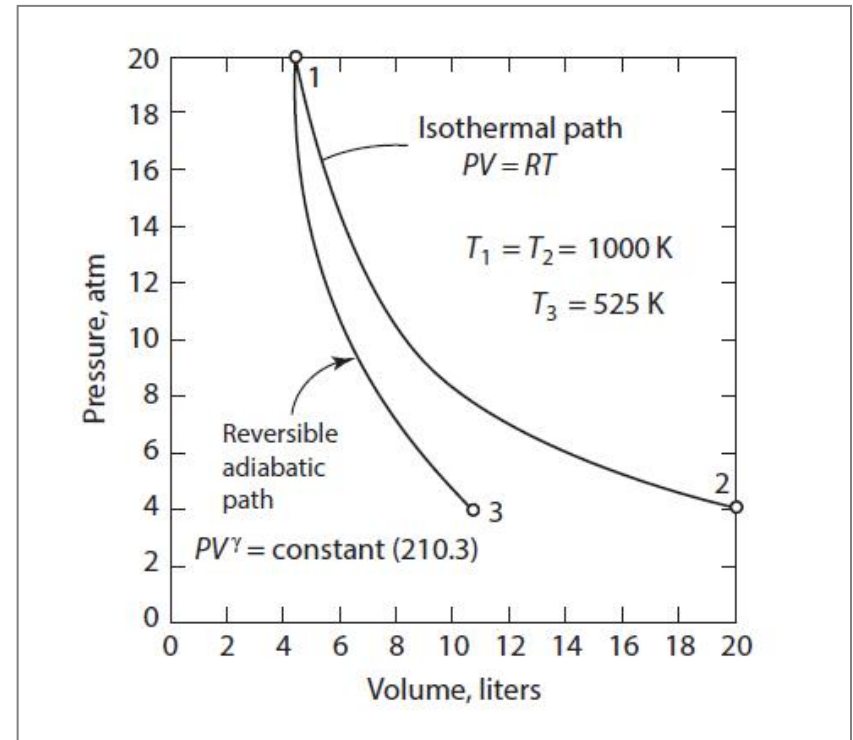


Fig. 2.4. Comparison of the process path taken by a reversible isothermal expansion of one mole an ideal gas with the process path taken by a reversible adiabatic expansion of an ideal gas between an initial pressure of 20 atm and a final pressure of 4 atm.

2.8 Reversible isothermal pressure or volume changes of an ideal gas

For the reversible isothermal path

$$w = q = RT \ln \frac{V_2}{V_1} = RT \ln \frac{P_1}{P_2}$$
$$= 8.3144 \times 1000 \times \ln \frac{20}{4} = 13.38 \text{ kJ}$$

For reversible adiabatic path:

$$PV^\gamma = \text{constant} = 210.3$$

$$w = \int P dV = 210.3 \times \int \frac{1}{V^{5/3}} dV$$

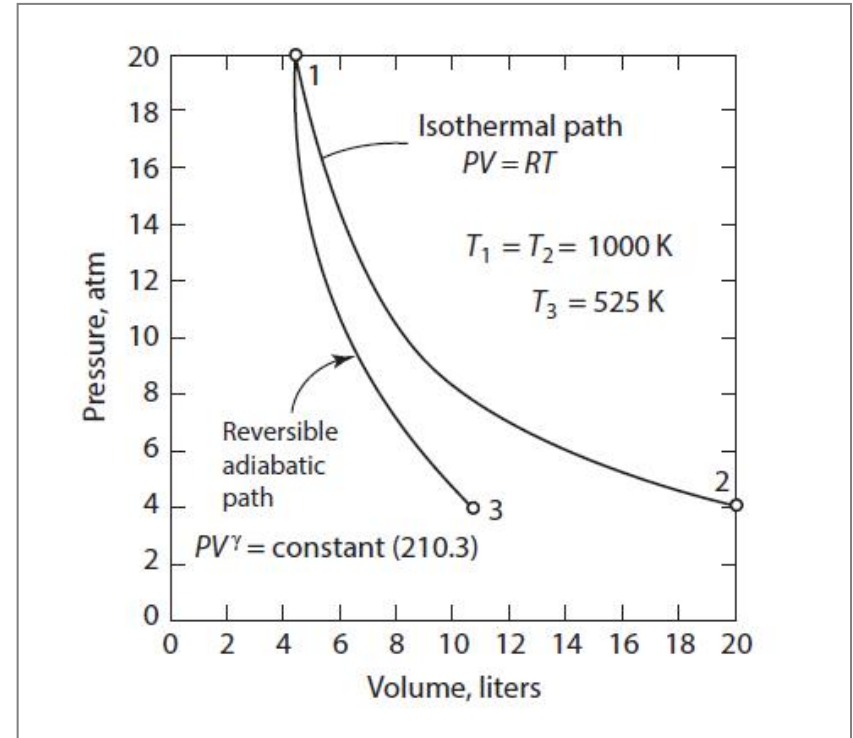


Fig. 2.4. Comparison of the process path taken by a reversible isothermal expansion of one mole an ideal gas with the process path taken by a reversible adiabatic expansion of an ideal gas between an initial pressure of 20 atm and a final pressure of 4 atm.

2.8 Reversible isothermal pressure or volume changes of an ideal gas

The volume of the gas in the state 3 is obtained as $V_3=(210.3/4)^{3/5} = 10.78$ liters, and the volume in the state 1 is $V_1=(210.3/20)^{3/5} = 4.10$ liters.

$$\begin{aligned} w &= \int P dV = 210.3 \times \int \frac{1}{V^{5/3}} dV \\ &= 210.3 \times \left(-\frac{3}{2} \right) V^{-2/3} \Big|_{V_1}^{V_3} = 5.92 \text{ kJ} \end{aligned}$$

Alternatively, since $q = 0$,

$$w = -\Delta U = c_v (T_3 - T_1) = -1.5 \times 8.3144 \times (525 - 1000) = 5.92 \text{ kJ}.$$

More work is performed during the isothermal process than during the adiabatic process.

2.9 Other forms of work

Up to this point, we have only used PdV work in the First Law. We called these systems *simple systems*. There are, of course, other kinds of work that may be performed on or by a system. Here, we present three additional kinds of work that we will utilize. To include other forms of work, the First Law can be written as

$$dU = \delta q - \sum \delta w_i$$

It is important to remember that in this equation, each of the terms δw_i is the work done by the system.

2.9 Electrical Work on a Dielectric Material

The work done on a dielectric material by an external electric field is given by:

$$\delta w = -V \mathbf{E} d\mathbf{D}$$

$$w = -V \int_{D_1}^{D_2} \mathbf{E} d\mathbf{D}$$

$\mathbf{E}$ is the electric field intensity (N/coul) and $\mathbf{D}$ is electric displacement (coul·m/m³). The units of work are in joules. Both $\mathbf{E}$ and $\mathbf{D}$ are *polar vectors*

The first law: $dU = \delta q - \sum \delta w_i = \delta q - PdV + V\mathbf{E}d\mathbf{D}$

For an adiabatic process at constant volume

$$(dU)_{q,V} = \delta q - \sum \delta w_i = V\mathbf{E}d\mathbf{D}$$

Under the specified conditions, the internal energy, U , of the dielectric material increases if an applied electric field increases its displacement field.

2.9 Magnetic Work on a Paramagnetic Material

The work done *on* a material by an external magnetic field is given by

$$\delta w = -V \mu_0 \mathfrak{H} d\mathbf{M}$$
$$\delta w = -V \mu_0 \int_{M_1}^{M_2} \mathfrak{H} d\mathbf{M}$$


Axial vectors

$\mathbf{M}$ and $\mathfrak{H}$ have units of amp/m, μ_0 the permeability of vacuum has units of N/amp², and V is in units of m³.

The first law: $dU = \delta q - \sum \delta w_i = \delta q - PdV + V \mu_0 \mathfrak{H} d\mathbf{M}$

For an adiabatic process at constant volume

$$(dU)_{q,V} = \delta q - \sum \delta w_i = V \mu_0 \mathfrak{H} d\mathbf{M}$$

Under the specified conditions, the internal energy, U , of the paramagnetic material increases when an applied magnetic field increases its magnetization.

2.9 Work to Create or Extend a Surface

The reversible work done in *creating* a new surface (in joules) of area A is given as

$$\delta w = -\gamma dA$$

γ having units of J/m² and A having units of m².

The surface energy, γ , is a scalar. For liquids, the surface energy is isotropic; for solids, the surface energy has a *symmetry* that is at least that of the point symmetry of the solid.

The first law:
$$dU = \delta q - \sum \delta w_i = \delta q - PdV + \gamma dA$$

For an adiabatic process at constant volume

$$(dU)_{q,V} = \gamma dA$$

Under the specified conditions, the total energy of the system increases when new surface area is created.

2.9 Work to Create or Extend a Surface

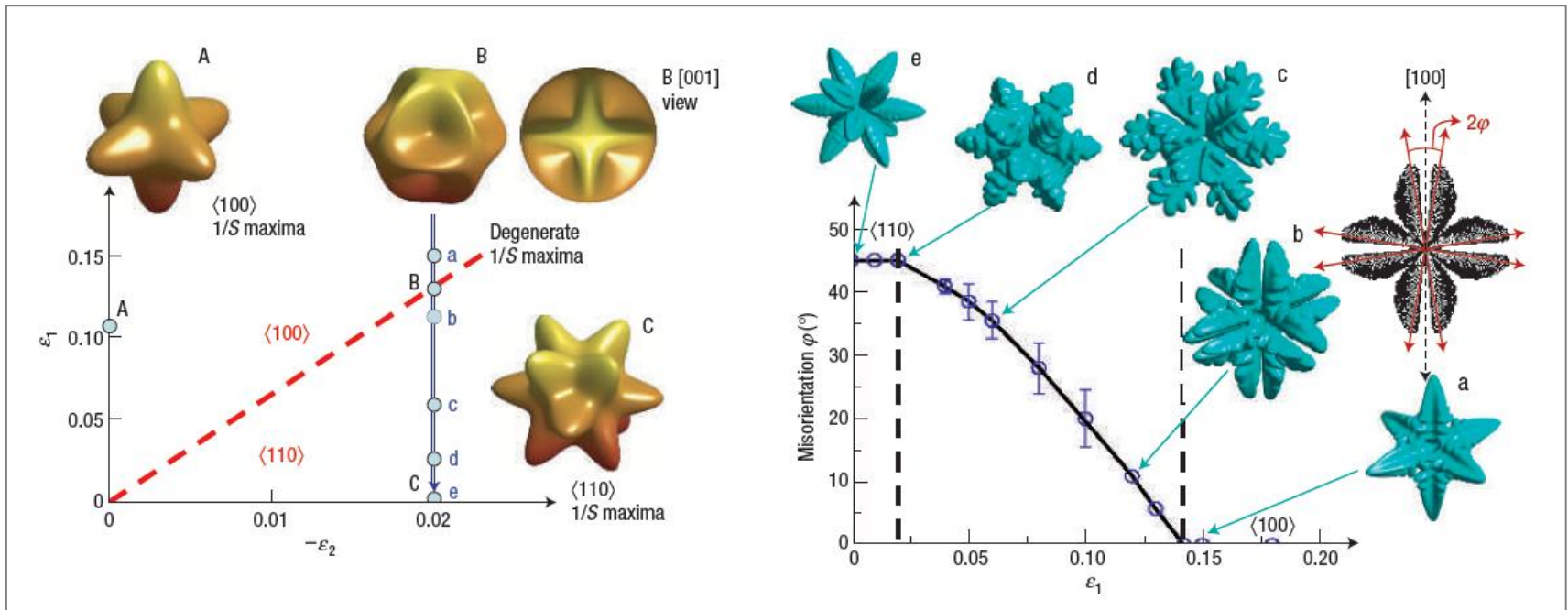
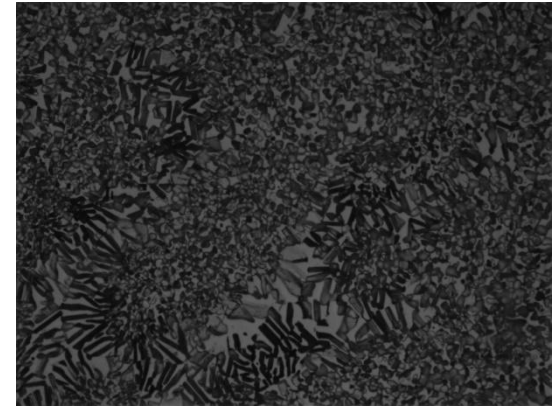
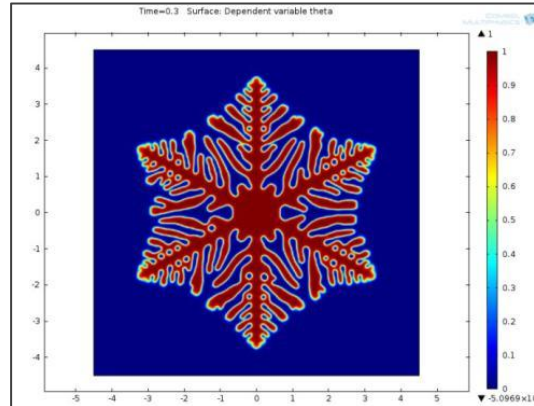
The volume terms are much larger than the surface terms, and hence, the surface terms may be neglected. Surface terms need to be included when the *surface-to-volume ratio* is not negligible, such as is the case for small particles.

A related quantity of work is the reversible work necessary to *increase* the surface area by *stretching* it with a stress σ . This work is given as

$$\delta w = -\sigma dA$$

Surface stress is a second-rank tensor, with units of N/m. For a liquid, the surface stress, σ , is isotropic and is numerically equal to the surface energy, γ , which is positive. *This is not necessarily the case for crystalline solids, where the surface stress may take on positive or negative values.*

2.9 Work to Create or Extend a Surface



2.12 Qualitative Example Problems

1. One mole of an ideal gas at a pressure of 1 atm and temperature of 273 K is expanded to twice its volume at constant pressure.
 - a.** In terms of P and V_1 and V_2 , how much work was performed? Was the work performed on the gas or by the gas? Explain.
 - b.** If two moles of an ideal gas at the same initial pressure and temperature were to double its volume under constant pressure, how much work would be performed on or by the gas? Compare the value with that in part (a).
 - c.** In either case (a) or (b), does the temperature increase or decrease? Explain.

Solution:

- a.** The work is performed *by* the gas and equals $P\Delta V = P(V_2 - V_1) = PV_1$.
- b.** The starting volume for this part of the problem is $2V_1 = V_2$. This doubles to $4V_1$. Thus, the work performed by 2 moles of the gas is $2PV_1$, twice that of part (a) of the problem.
- c.** The temperature increases in both cases. Take case (a): $P_1V_1 = RT_1$ and $P_2V_2 = RT_2$, but $P_2 = P_1$ and $V_2 = 2V_1$. Thus, $T_2 = 2T_1$. The temperature increases.

2.13 Qualitative Example Problems

2. Ten liters of a monatomic ideal gas at 25 °C and 10 atm pressure are expanded to a final pressure of 1 atm. The molar heat capacity of the gas at constant volume, c_v , is $3/2R$ and is independent of temperature. Calculate the work done, the heat absorbed, and the change in U and in H for the gas if the process is carried out.

- Isothermally and reversibly
- Adiabatically and reversibly

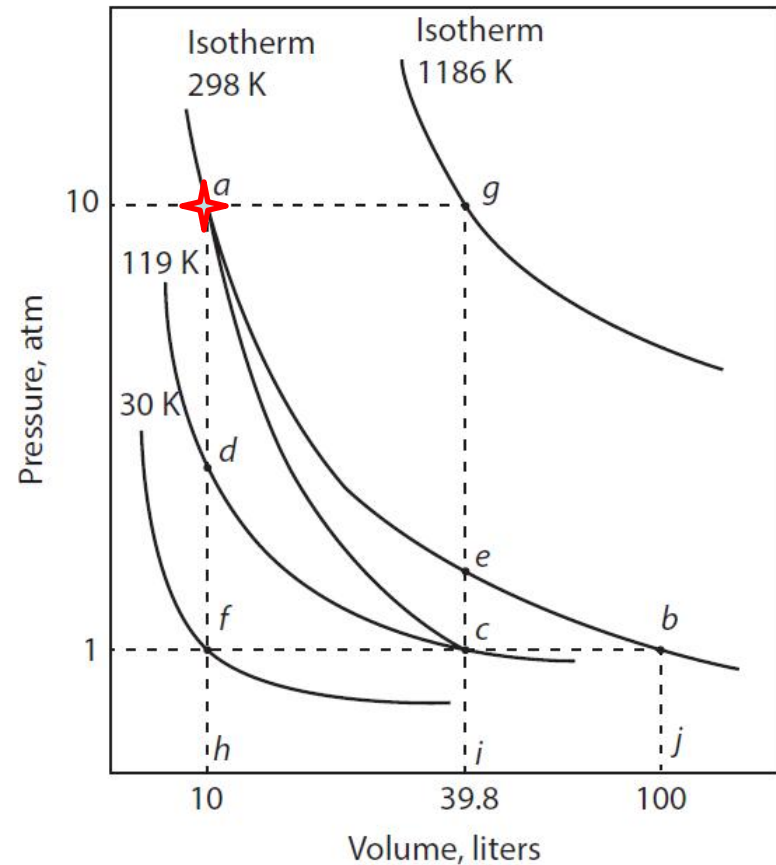
2.13 Qualitative Example Problems

Having determined the final state of the gas after the reversible adiabatic expansion, verify that the change in U for the process is independent of the path taken between the initial and final states by considering the process to be carried out as

- An isothermal process followed by a constant-volume process
- A constant-volume process followed by an isothermal process
- An isothermal process followed by a constant-pressure process
- A constant-volume process followed by a constant-pressure process
- A constant-pressure process followed by a constant-volume process

2.13 Qualitative Example Problems

From consideration of the initial state of the system (the point **a**),



$$n = \text{the number of moles} = \frac{P_a V_a}{RT_a} = \frac{10 \times 10}{0.08206 \times 298} = 4.09$$

2.13 Qualitative Example Problems

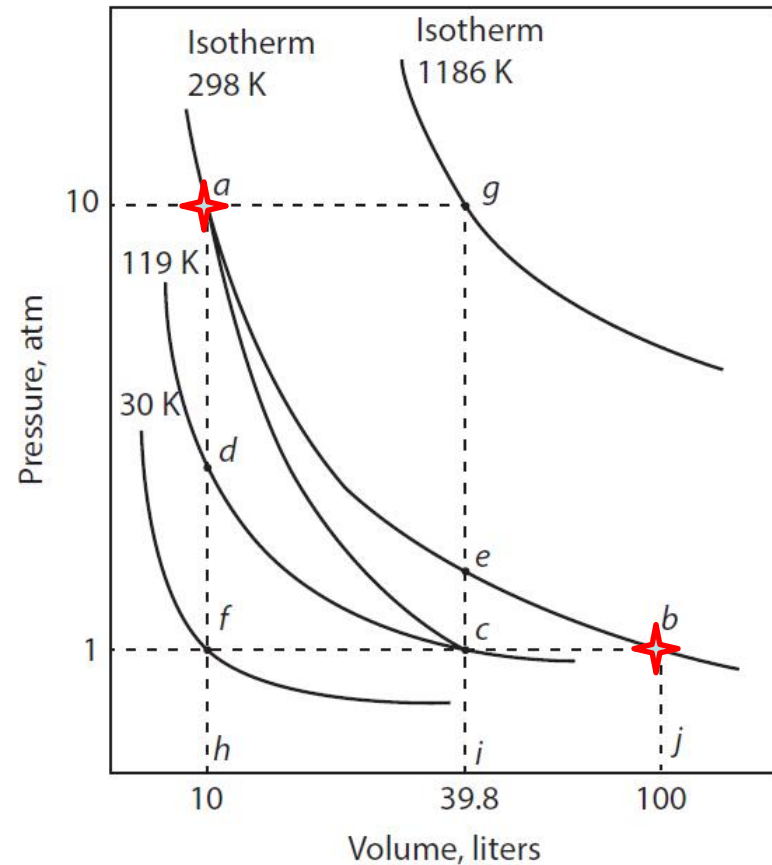
a. The isothermal reversible expansion

The state of the gas moves from *a* to *b* along the 298-degree isotherm. Along any isotherm, the product *PV* is constant.

$$V_b = \frac{P_a V_a}{P_b} = \frac{10 \times 10}{1} = 100 \text{ liters}$$

For an ideal gas undergoing an isothermal process, $\Delta U=0$, and hence, from the First Law,

$$q = w = \int_a^b P dV = nRT \int_a^b \frac{1}{V} dV = 4.09 \times 8.3144 \times 298 \times \ln \frac{100}{10} = 23.3 \text{ kJ}$$

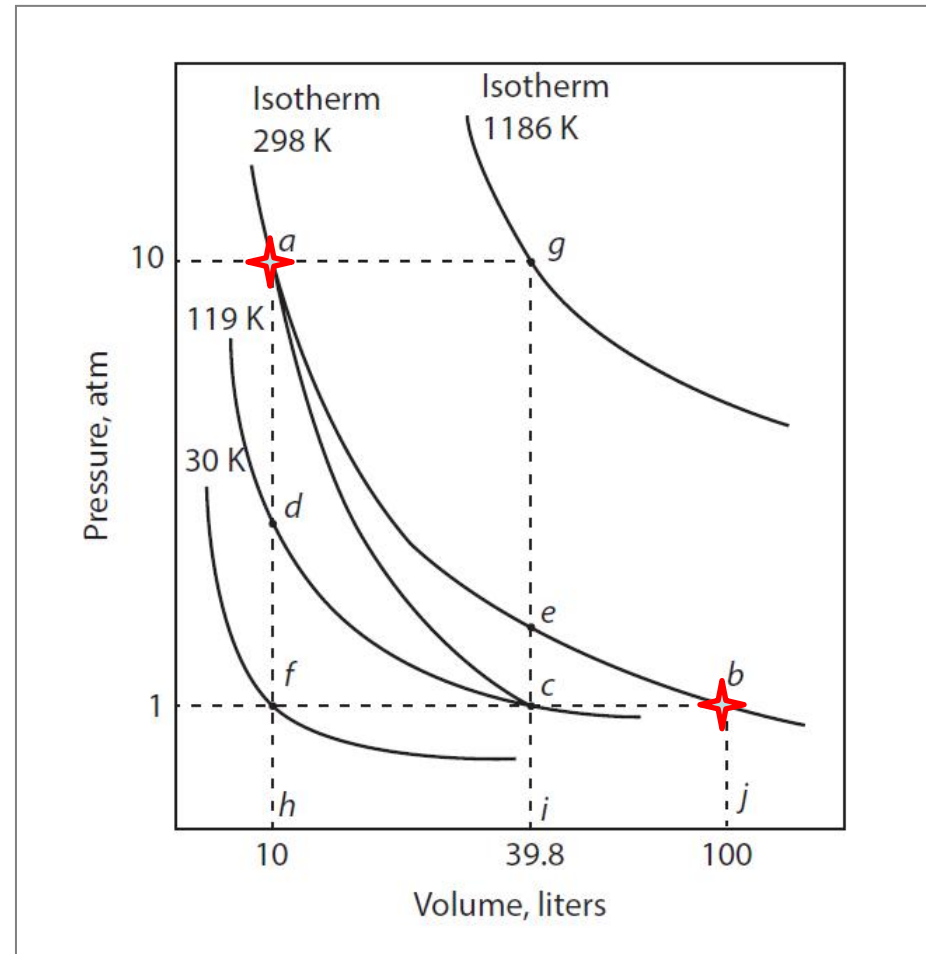


2.13 Qualitative Example Problems

In passing from the state **a** to the state **b** along the 298-degree isotherm, the system performs 23.3 kJ of work and absorbs 23.3 kJ of heat from the constant temperature surroundings.

Since for an ideal gas, H is a function only of temperature, ΔH ($a \rightarrow b$) = 0; that is,

$$\begin{aligned}\Delta H_{a \rightarrow b} &= \Delta U_{a \rightarrow b} + (P_b V_b - P_a V_a) \\ &= (P_b V_b - P_a V_a) \\ &= nRT_b - nRT_a \\ &= 0\end{aligned}$$

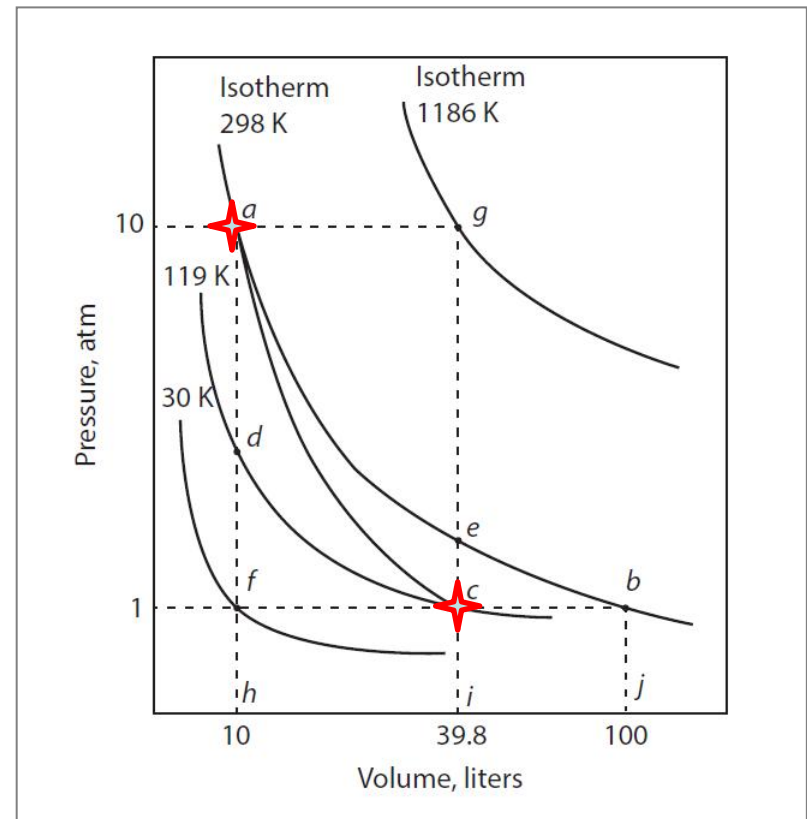


2.13 Qualitative Example Problems

b. The reversible adiabatic expansion. If the adiabatic expansion is carried out reversibly, then during the process the state of the system is, at all times, given by $PV^\gamma = \text{constant}$, and the final state is the point c in the diagram. The volume V_c is obtained from $P_a V_a^\gamma = P_c V_c^\gamma$ as

$$V_c = \left(\frac{10 \times 10^{5/3}}{1} \right)^{3/5} = 39.8 \text{ liters}$$

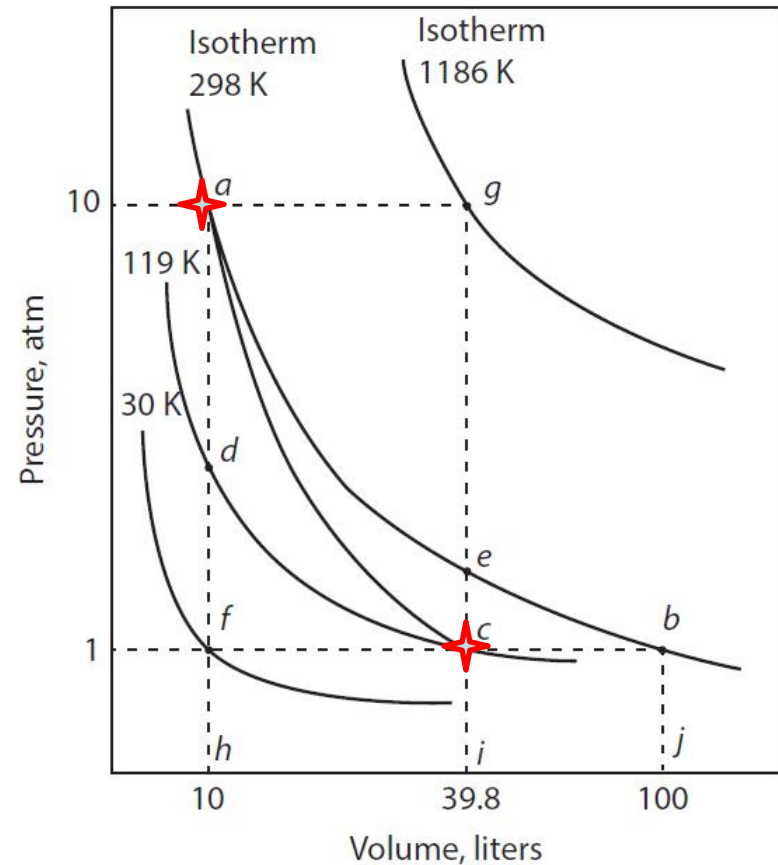
$$T_c = \frac{P_c V_c}{nR} = \frac{1 \times 39.8}{4.09 \times 0.08206} = 119 \text{ degrees}$$



2.13 Qualitative Example Problems

The point **c** thus lies on the 119-degree isotherm. As the process is adiabatic, $q = 0$, and hence,

$$\begin{aligned}\Delta U_{(a \rightarrow c)} &= -w = \int_a^c n c_V dT \\ &= n c_V (T_c - T_a) \\ &= 4.09 \times 1.5 \times 8.3144 \times (119 - 298) \\ &= -9.13 \text{ kJ}\end{aligned}$$



The work done by the system as a result of the process equals the decrease in the internal energy of the system = 9.13 kJ.

2.13 Qualitative Example Problems

i. **An isothermal process followed by a constant-volume process** (the path **a**→**e**→**c** ; that is, an isothermal change from a to e , followed by a constant-volume change from e to c).

- isothermal process

$$\Delta U_{(a \rightarrow e)} = 0$$

$$\Delta U_{(e \rightarrow c)} = \int_e^c n c_V dT$$

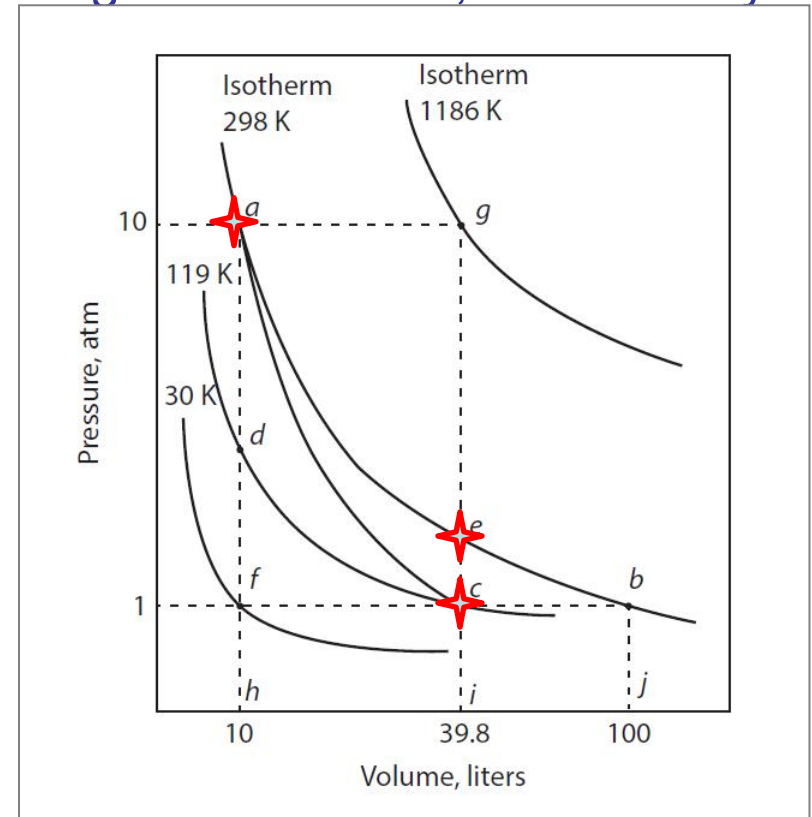
- isochoric process

$$\Delta V = 0 \Rightarrow w = 0$$

As the state e lies on the 298-degree isotherm, then

$$\Delta U_{(e \rightarrow c)} = 4.09 \times 1.5 \times 8.3144 \times (119 - 298) = -9.13 \text{ kJ}$$

$$\Delta U_{(a \rightarrow c)} = \Delta U_{(a \rightarrow e)} + \Delta U_{(e \rightarrow c)} = -9.13 \text{ kJ}$$



2.13 Qualitative Example Problems

ii. *A constant-volume process followed by an isothermal process* (the path $a \rightarrow d \rightarrow c$; that is, a constant-volume change from a to d , followed by an isothermal change from d to c).

- isochoric process

$$\Delta V = 0 \Rightarrow w = 0$$

The state d lies in the 119-degree isotherm

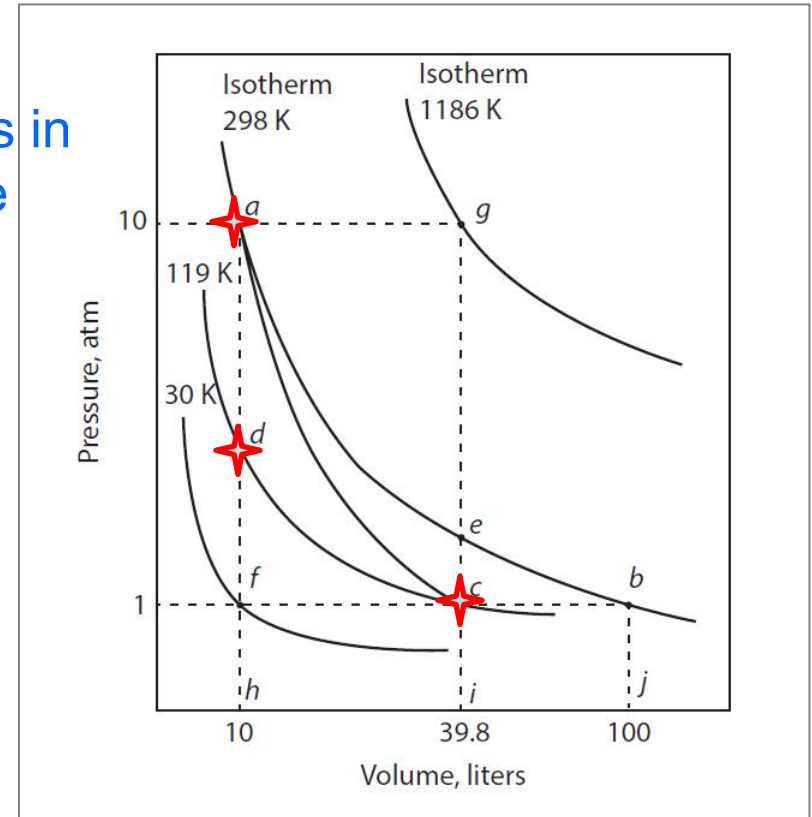
$$\begin{aligned}\Delta U_{(a \rightarrow d)} &= q_V = \int_e^c n c_V dT \\ &= 4.09 \times 1.5 \times 8.3144 \times (119 - 298) \\ &= -9.13 \text{ kJ}\end{aligned}$$

- isothermal process

$$\Delta U_{(d \rightarrow c)} = 0$$

$$\Delta U_{(a \rightarrow c)} = \Delta U_{(a \rightarrow d)} + \Delta U_{(d \rightarrow c)} = -9.13 \text{ kJ}$$

Thermodynamics of Materials & Phase transformations



2.13 Qualitative Example Problems

iii. *An isothermal process followed by a constant-pressure process* (the path $a \rightarrow b \rightarrow c$; that is, an isothermal change from a to b , followed by a constant pressure change from b to c).

- isothermal process

$$\Delta U_{(a \rightarrow b)} = 0$$

- isobaric process

$$\Delta U_{(b \rightarrow c)} = q_p - w$$

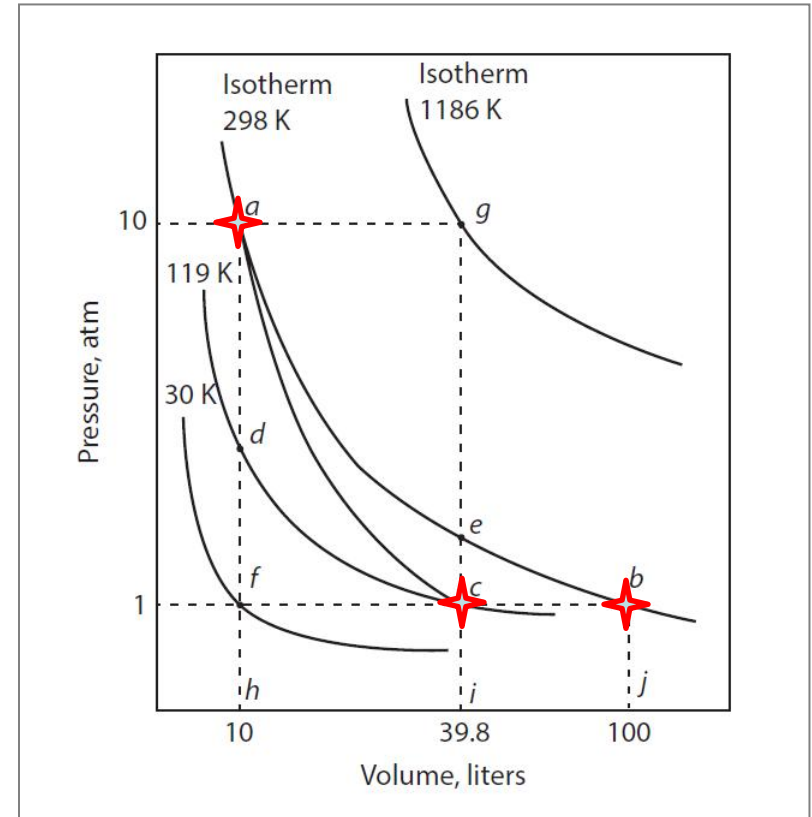
$$= \int_b^c n c_P dT - P_b (V_c - V_b)$$

$$= 4.09 \times 2.5 \times 8.3144 \times (119 - 298)$$

$$- (39.8 - 100) \times 101.3$$

$$= -9.13 \text{ kJ}$$

$$\Delta U_{(a \rightarrow c)} = \Delta U_{(a \rightarrow b)} + \Delta U_{(b \rightarrow c)} = -9.13 \text{ kJ}$$



2.13 Qualitative Example Problems

iv. A **constant-volume process followed by a constant-pressure process** (the path **a**→**f**→**c**; that is, a constant-volume change from **a** to **f**, followed by a constant-pressure change from **f** to **c**).

- isochoric process

$$\Delta U_{(a \rightarrow f)} = q_V = \int_a^f n c_V dT$$

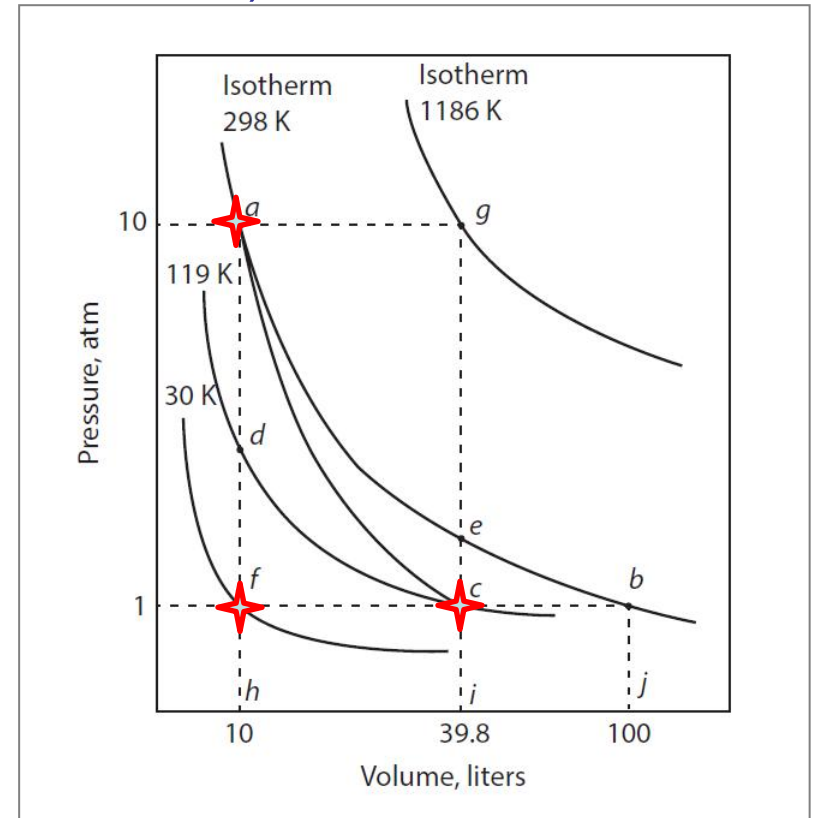
$$T_f = \frac{P_f V_f}{nR} = \frac{1 \times 10}{4.09 \times 0.08206}$$

$$= 30 \text{ degrees}$$

$$\Delta U_{(a \rightarrow f)}$$

$$= 4.09 \times 1.5 \times 8.3144 \times (30 - 298)$$

$$= -13.67 \text{ kJ}$$



2.13 Qualitative Example Problems

iv. **A constant-volume process followed by a constant-pressure process** (the path **a**→**f**→**c**; that is, a constant-volume change from *a* to *f*, followed by a constant-pressure change from *f* to *c*).

- isobaric process

$$\Delta U_{(f \rightarrow c)} = q_p - w$$

$$= \int_f^c n c_p dT - P_f (V_c - V_f)$$

$$= 4.09 \times 2.5 \times 8.3144 \times (119 - 30) - (39.8 - 10) \times 101.3$$

$$= 7.57 - 3.02 \text{ kJ}$$

$$\Delta U_{(a \rightarrow c)} = \Delta U_{(a \rightarrow f)} + \Delta U_{(f \rightarrow c)} = -9.13 \text{ kJ}$$

2.13 Qualitative Example Problems

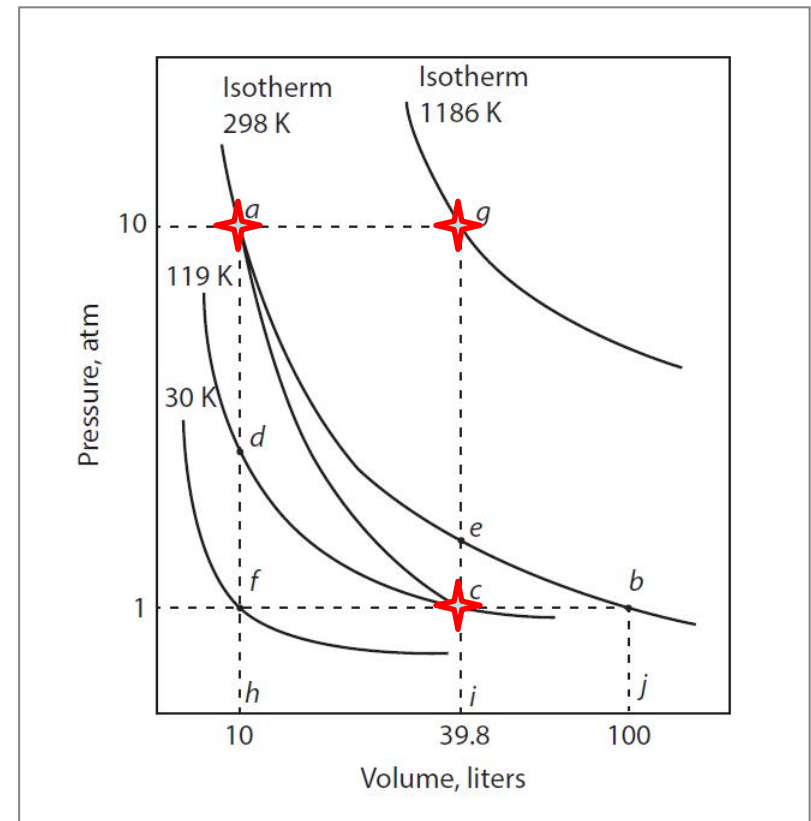
v. A constant-pressure process followed by a constant-volume process (the path $a \rightarrow g \rightarrow c$; that is, a constant-pressure step from a to g , followed by a constant volume step from g to c).

- isobaric process

$$\Delta U_{(a \rightarrow g)} = q_p - w$$

$$T_g = \frac{P_g V_g}{nR}$$

$$= \frac{10 \times 39.8}{4.09 \times 0.08206} = 1186 \text{ degrees}$$



2.13 Qualitative Example Problems

v. A constant-pressure process followed by a constant-volume process (the path $a \rightarrow g \rightarrow c$; that is, a constant-pressure step from a to g , followed by a constant volume step from g to c).

$$\begin{aligned}\Delta U_{(a \rightarrow g)} &= 4.09 \times 2.5 \times 8.3144 \times (1186 - 298) - 10 \times (39.8 - 10) \times 101.3 \\ &= 75.5 - 30.2\end{aligned}$$

● isochoric process

$$\begin{aligned}\Delta U_{(g \rightarrow c)} &= q_V \\ &= 4.09 \times 1.5 \times 8.3144 \times (119 - 1186) \\ &= -54.4 \text{ kJ}\end{aligned}$$

$$\Delta U_{(a \rightarrow c)} = \Delta U_{(a \rightarrow g)} + \Delta U_{(g \rightarrow c)} = -9.13 \text{ kJ}$$

2.13 Qualitative Example Problems

The change in enthalpy from a to c

The enthalpy change is most simply calculated from the consideration of a path which involves an isothermal portion, over which $\Delta H' = 0$, and an isobaric portion, over which $\Delta H' = q_p = \int n c_p dT$. For example, consider the path $a \rightarrow b \rightarrow c$.

$$\Delta H_{(a \rightarrow b)} = 0$$

$$\begin{aligned}\Delta H_{(b \rightarrow c)} &= q_P = n c_P (T_c - T_b) = 4.09 \times 2.5 \times 8.3144 \times (119 - 298) \\ &= -15.2 \text{ kJ}\end{aligned}$$

$$\Delta H_{(a \rightarrow c)} = \Delta H_{(a \rightarrow b)} + \Delta H_{(b \rightarrow c)} = -15.2 \text{ kJ}$$

Alternatively
$$\begin{aligned}\Delta H_{(a \rightarrow c)} &= \Delta U_{(a \rightarrow c)} + P_c V_c - P_a V_a \\ &= 9.12 + (1 \times 39.8 - 10 \times 10) \times 101.3 = -15.2 \text{ kJ}\end{aligned}$$

Calculations of q , w , ΔU and ΔH for various processes

□ Calculation of ΔU and ΔH for various physical processes of ideal gases

The internal energy and enthalpy of a certain amount of ideal gas are only functions of temperature, independent of pressure and volume, and their molar heat capacity at constant pressure and at constant volume are also not a function of temperature.

$$\Delta U = q_V = \int C_V dT = n c_{V,m} (T_2 - T_1) \quad \Delta H = q_p = \int C_p dT = n c_{p,m} (T_2 - T_1)$$

□ Calculation of volume work

$$\delta w = p_e dV \quad \left\{ \begin{array}{l} w = \int p_e dV = p_e (V_2 - V_1) \\ w = \int p dV \\ w = -\Delta U \end{array} \right.$$

□ Calculation of heat

Simple temperature change
process without phase
change and chemical change

$$q = \int C dT$$

Calculations of q , w , ΔU and ΔH for various processes

□ Formulas of q , w , ΔU , ΔH for various processes

Only change of P , V , T : ideal gas

Isothermal reversible process $\Delta U = \Delta H = 0$ $q = -w = nRT \ln \frac{V_2}{V_1}$

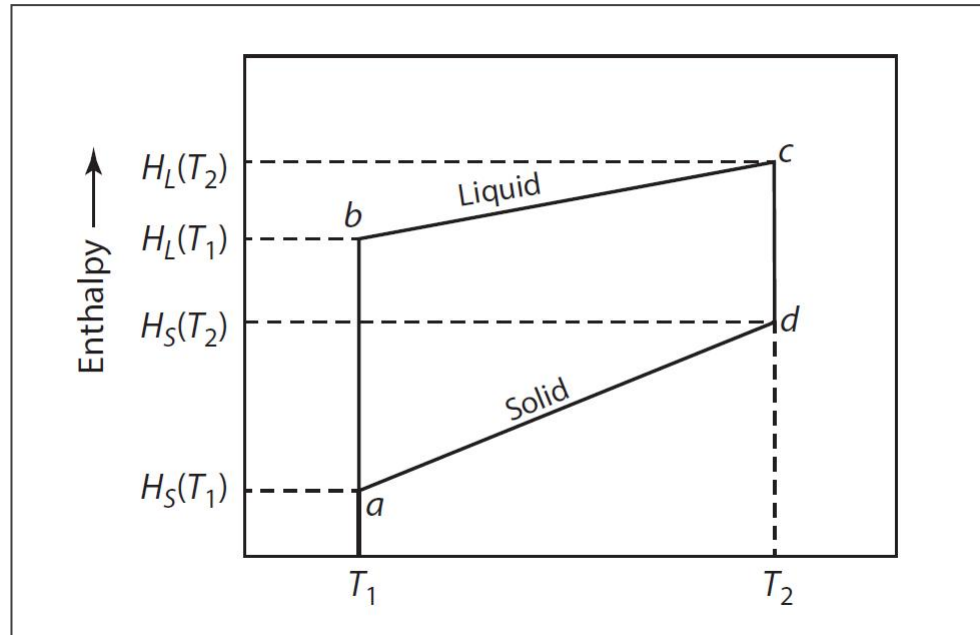
isobaric process $\Delta U = \int C_v dT$ $Q = \Delta H = \int C_p dT$ $w = P(V_2 - V_1)$

isochoric process $\Delta U = \int C_v dT$ $\Delta H = \int C_p dT$ $q = \Delta U$ $w = 0$

adiabatic process $\Delta U = \int C_v dT$ $\Delta H = \int C_p dT$ $q = 0$ $w = -\Delta U$

Calculations of q , w , ΔU and ΔH for various processes

Phase-transformations



Chemical reactions

Homework

2.1 An monatomic ideal gas at 300 K has a volume of 15 liters at a pressure of 15 atm.

Calculate

- a. The final volume of the system
- b. The work done by the system
- c. The heat entering or leaving the system
- d. The change in the internal energy
- e. The change in the enthalpy when the gas undergoes
 - i. A reversible isothermal expansion to a pressure of 10 atm
 - ii. A reversible adiabatic expansion to a pressure of 10 atm

The constant-volume molar heat capacity of the gas, c_v , has the value $1.5 R$.

2.3 The initial state of a quantity of monatomic ideal gas is $P=1$ atm, $V=1$ liter, and $T=373$ K. The gas is isothermally expanded to a volume of 2 liters and is then cooled at constant pressure to the volume V . This volume is such that a reversible adiabatic compression to a pressure of 1 atm returns the system to its initial state. All of the changes of state are conducted reversibly. Calculate the value of V and the total work done on or by the gas.

Homework

2.5 One mole of N_2 gas is contained at 273 K and a pressure of 1 atm. The addition of 3000 J of heat to the gas at constant pressure causes 832 J of work to be done during the expansion. Calculate

- The final state of the gas
- The values of ΔU and ΔH for the change of state
- The values of c_v and c_p for N_2

Assume that nitrogen behaves as an ideal gas, and that the change of state is conducted reversibly.

2.7 One mole of an monatomic ideal gas at 25 °C and 1 atm undergoes the following reversibly conducted cycle:

- An isothermal expansion to 0.5 atm, followed by
- An isobaric expansion to 100 °C, followed by
- An isothermal compression to 1 atm, followed by
- An isobaric compression to 25 °C

The system then undergoes the following reversible cyclic process:

- An isobaric expansion to 100 °C, followed by
- A decrease in pressure at constant volume to the pressure P atm, followed by
- An isobaric compression at P atm to 24.5 liters, followed by
- An increase in pressure at constant volume to 1 atm

Calculate the value of P which makes the work done on the gas during the first cycle equal to the work done by the gas during the second cycle.

Results

- (1) Reversible expansion of an ideal gas at a constant temperature (volume increases), W_{00} , Q_{00} , ΔU_{00} , ΔH_{00} ;
- (2) Adiabatic throttling expansion of an ideal gas, W_{00} , Q_{00} , ΔU_{00} , ΔH_{00} ;
- (3) Isobaric expansion of an ideal gas ($\Delta V > 0$), W_{00} , Q_{00} , ΔU_{00} , ΔH_{00} ;
- (4) Free expansion of an ideal gas at constant temperature, W_{00} , Q_{00} , ΔU_{00} , $\Delta H=0$;
- (5) Adiabatic free expansion of a real gas, W_{00} , Q_{00} , ΔU_{00} , ΔT_{00} ;
- (6) Free expansion of a real gas under a constant temperature, W_{00} , Q_{00} , ΔU_{00} ;
- (7) The ice melts into water. Taking ice and water as the system, Q_{00} , W_{00} , ΔU_{00} , ΔH_{00} ;
- (8) Water vapor does some work by the steam engine and returns to the initial state. Taking the water vapor as the system, Q_{00} , W_{00} , ΔU_{00} , ΔH_{00}

Results

- (1) Reversible expansion of an ideal gas at a constant temperature (volume increases),
 $W > 0$, $Q > 0$, $\Delta U = 0$, $\Delta H = 0$;
- (2) Adiabatic throttling expansion of an ideal gas, $W = 0$, $Q = 0$, $\Delta U = 0$, $\Delta H = 0$;
- (3) Isobaric expansion of an ideal gas ($\Delta V > 0$), $W > 0$, $Q > 0$, $\Delta U > 0$, $\Delta H > 0$;
- (4) Free expansion of an ideal gas at constant temperature, $W = 0$, $Q = 0$, $\Delta U = 0$,
 $\Delta H = 0$;
- (5) Adiabatic free expansion of a real gas, $W = 0$, $Q = 0$, $\Delta U = 0$, $\Delta T \neq 0$;
- (6) Free expansion of a real gas under a constant temperature, $W = 0$, $Q > 0$, $\Delta U > 0$;
- (7) The ice melts into water. Taking ice and water as the system, $Q > 0$, $W < 0$, $\Delta U > 0$,
 $\Delta H > 0$;
- (8) Water vapor does some work by the steam engine and returns to the initial state.
Taking the water vapor as the system, $Q > 0$, $W > 0$, $\Delta U = 0$, $\Delta H = 0$

Thermodynamics of Materials & Phase transformations

The End



2.6 Heat capacity

$$dU = \left(\frac{\partial U}{\partial T} \right)_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$



Divided by dT

$$\frac{dU}{dT} = \left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \frac{dV}{dT}$$



$$\left(\frac{\partial U}{\partial T} \right)_P = \left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P$$

2.6 Heat capacity

Prove:
$$\left(\frac{\partial U}{\partial T}\right)_P = \left(\frac{\partial U}{\partial T}\right)_V + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P$$

Suppose: $U = U(T, V), \quad V = V(T, P)$

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

$$dV = \left(\frac{\partial V}{\partial T}\right)_P dT + \left(\frac{\partial V}{\partial P}\right)_T dP$$

Substituting the expression of dV , we have:

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T \left[\left(\frac{\partial V}{\partial T}\right)_P dT + \left(\frac{\partial V}{\partial P}\right)_T dP \right]$$

2.6 Heat capacity

Rearranging to separate dP from dT , we have:

$$\begin{aligned} dU &= \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial P} \right)_T dP + \left[\left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P \right] dT \\ &= \left(\frac{\partial U}{\partial P} \right)_T dP + \left[\left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P \right] dT \end{aligned}$$

Because U is a function of T and P . $U = U(T, P)$

$$dU = \left(\frac{\partial U}{\partial P} \right)_T dP + \left(\frac{\partial U}{\partial T} \right)_P dT$$

According to the two expression of dU , we have:

$$\left(\frac{\partial U}{\partial T} \right)_P = \left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P$$